

Assignment 2

Exploring Public Single-Cell & Spatial Omics Data Repositories

Objective

Become familiar with major public repositories used for single-cell and spatial omics research. Learn how datasets can be searched, explored, downloaded, and in some cases analyzed directly through the repository website.

Task

Each student should choose TWO repositories from the list below and prepare a short presentation for the next session.

Repositories

- 10x Genomics Datasets
- GEO / NCBI
- CELLxGENE
- Human Cell Atlas
- SpatialDB
- STOmicsDB
- Broad Single Cell Portal
- ArrayExpress / BioStudies

What to Investigate

- Repository overview (purpose, organization, available data types)
- Available datasets (single-cell, spatial transcriptomics, multiomics, etc.)
- Dataset search process
- Dataset download process
- Available file formats
- Online analysis or visualization features (if available)
- Strengths and limitations
- Ease of use for new researchers

Practical Demonstration

- Find at least one real dataset in each repository.
- Show how the dataset can be located.
- Demonstrate the download process.
- If analysis tools are available, demonstrate their usage.
- Take screenshots of important steps.

Presentation Requirements

- Repository introduction

- Available data types
- Dataset search process
- Dataset download process
- Available analysis and visualization features
- Screenshots from your exploration
- Strengths and limitations
- Final recommendation

Live Demonstration

- Open the repository website.
- Show how a dataset is found.
- Demonstrate the download process.
- Demonstrate any available analysis or visualization tools.

Final Question

Which repository would you choose for a new Single-Cell or Spatial Omics project and why?

Special Note

Repositories such as CELLxGENE, STOmicsDB, and Broad Single Cell Portal provide interactive visualization and exploration tools in addition to dataset downloads. If one of your selected repositories includes online analysis capabilities, be sure to demonstrate these features during your presentation.